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Systems and methods for real-time accent mimicking

Abstract

The disclosed technology relates to methods, speech processing systems, and non-transitory computer readable media for real-time accent mimicking. In some examples, trained machine learning model(s) are applied to first input audio data to extract accent features of first input speech associated with a first accent of a first user. Obtained second input data associated with second input speech associated with a second accent of a second user is analyzed to generate characteristics specific to a natural voice of the second user. A modified version of the second input speech is synthesized based on the generated characteristics and the extracted accent features. The modified version of the second input speech advantageously preserves aspects of the natural voice of the second user and mimics the first accent. Output audio data generated based on the modified version of the second input speech is provided for output via an audio output device.

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Background/Summary

(1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/678,180, filed Aug. 1, 2024, which is hereby incorporated herein by reference in its entirety.

FIELD

(1) This technology generally relates to audio analysis and, more particularly, to methods and systems for real-time accent mimicking.

BACKGROUND

(2) Effective communication is a fundamental aspect of human interaction, essential for personal, educational, and professional success. Clarity and understandability of speech are critical to enable speakers to convey their thoughts and listeners to comprehend the intended message accurately. However, regional accents can often create barriers to understanding, especially for individuals who are not familiar with a particular dialect. These barriers can lead to misunderstandings, reduced efficiency in communication, and even social and professional disadvantages.

(3) Language learners, in particular, face significant challenges related to accents. The nuances of pronunciation, intonation, and rhythm in a target language's accent can be difficult to master. Learners often struggle to replicate these nuances, which can hinder their overall pronunciation development. Poor accent replication also can lead to difficulties in being understood by native speakers, impacting the learner's confidence and progression in the language.

(4) Existing technologies have attempted to address accent-related issues through various means. Some of these current technologies focus on static accent conversion, where a user's speech is transformed into a different accent using pre-programmed algorithms. While these static approaches offer some benefits, they lack the dynamic nature required for real-time interactions. Static conversion often results in unnatural-sounding speech and fails to adapt to the changing context of conversations.

(5) In response to the growing need for improved communication clarity, virtual conference platforms have begun incorporating accent conversion features. These solutions typically rely on machine learning models trained on prerecorded speech data. While this method can enhance understanding to some extent, it struggles to adapt to the nuances and variability of real-time conversations. The reliance on prerecorded data means that these machine learning models may not accurately capture the dynamic features of spontaneous speech, leading to potential inaccuracies and a lack of naturalness in the converted speech.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosed technology is illustrated by way of example and not limitation in the accompanying figures, in which like references indicate similar elements:

(2) FIG. 1 is a block diagram of an exemplary network environment that includes a speech processing system;

(3) FIG. 2 is a block diagram of an exemplary storage device of the speech processing system of FIG. 1;

(4) FIG. 3 is a flow diagram of an exemplary method for real-time accent mimicking; and
(5) FIG. 4 is a flowchart of an exemplary method for real-time accent mimicking.

DETAILED DESCRIPTION

(6) Examples described below may be used to provide a method, a device (e.g., non-transitory computer readable medium), an apparatus, and/or a system for real-time accent mimicking. Although the technology has been described with reference to specific examples, various modifications may be made to these examples without departing from the broader spirit and scope of the various embodiments of the technology described and illustrated by way of the examples herein. The disclosed technology includes a speech processing system **100** that aids speakers with accents in adopting listeners' accents, thereby enhancing communication clarity and reducing accent-related barriers, among other advantages explained in detail below.

(7) Referring now to FIG. 1, a block diagram of an exemplary network environment that includes a speech processing system **100** is illustrated. The speech processing system **100** in this example includes processor(s) **104**, which are designed to process instructions (e.g., computer readable instructions (i.e., code)) stored on the storage device(s) **114** (e.g., a non-transitory computer readable medium) of the speech processing system **100**. By processing the stored instructions, the processor(s) **104** may perform the steps and functions disclosed herein, such as with reference to FIG. 3-4, for example.

(8) The speech processing system **100** also includes an operating system and microinstruction code in some examples, one or both of which can be hosted by the storage device(s) **114**. The various processes and functions described herein may either be part of the microinstruction code and/or program code (or a combination thereof), which is executed via the operating system. The speech processing system **100** also may have data storage **106**, which along with the processor(s) **104** form a central processing unit (CPU) **102**, an input controller **110**, an output controller **112**, and/or a communication controller **108**. A bus **113** may operatively couple components of the speech processing system **100**, including processor(s) **104**, data storage **106**, storage device(s) **114**, input controller **110**, output controller **112**, and/or any other devices (e.g., a network controller or a sound controller).

(9) The output controller **112** may be operatively coupled (e.g., via a wired or wireless connection) to a display device (e.g., a monitor, television, mobile device screen, touch-display, etc.) in such a fashion that the output controller **112** can transform the display on the display device (e.g., a response to the execution of module(s)). Input controller **110** may be operatively coupled (e.g., via a wired or wireless connection) to an input device (e.g., mouse, keyboard, touchpad scroll-ball, touch-display, etc.) in such a fashion that input can be received from a user of the speech processing system **100**.

(10) The communication controller **108** in some examples provides a two-way coupling through a network link to the Internet **120** that is connected to a local network **118** and operated by an Internet service provider (ISP) **122**, which provides data communication services to the Internet **120**. The network link typically provides data communication through one or more networks to other data devices. For example, the network link may provide a connection through local network **118** to a host computer and/or to data equipment operated by the ISP **122**. A server **124** may transmit requested code for an application through the Internet **120**, ISP **122**, local network **118**, and/or communication controller **108**.

(11) The audio interface **126**, also referred to as a sound card, includes sound processing hardware and/or software, including a digital-to-analog converter (DAC) and an analog-to-digital converter (ADC). The audio interface **126** is coupled to a physical microphone **128** and an audio output device **130** (e.g., headphones or speaker(s)) in this example, although the audio interface **126** can be coupled to other types of audio devices in other examples. Thus, the audio interface **126** uses the ADC to digitize input analog audio signals from a sound source (e.g., the physical microphone **128**) so that the digitized signals can be processed by the speech processing system **100**, such as

according to the methods described and illustrated herein. The DAC of the audio interface **126** can convert generated digital audio data into an analog format for output via the audio output device **130**.

(12) The speech processing system **100** is illustrated in FIG. **1** with all components as separate devices for ease of identification only. One or more of the components of the speech processing system **100** in other examples may be separate devices (e.g., a personal computer connected by wires to a monitor and mouse), may be integrated in a single device (e.g., a mobile device with a touch-display, such as a smartphone or a tablet), or any combination of devices (e.g., a computing device operatively coupled to a touch-screen display device, a plurality of computing devices attached to a single display device and input device, etc.). The speech processing system **100** also may be one or more servers, for example a farm of networked or distributed servers, a clustered server environment, or a cloud.

(13) Referring now to FIG. **2**, a block diagram of an exemplary one of the storage device(s) **114** of the speech processing system **100** is illustrated. The storage device **114** may include an accent analysis module **200**, a natural speech preservation module **202**, an input interface **204**, an accent translation module **206**, an output module **208**, a synthesizer module **210**, and/or a feature extraction module **212**, although other types and/or number of modules can also be used in other examples.

(14) The input interface **204** may serve as an interface through which the speech processing system **100** receives input data and may allow for the input of speech and/or audio data or any other representation that captures characteristics of input speech. The input interface **204** may include various components or functionalities to facilitate the input process and may include hardware components such as microphones or audio interfaces for capturing real-time speech data. Alternatively, the input interface **204** may include a software interface that allows for the input of prerecorded speech data or textual representations, and other types of input interfaces can also be used in other examples.

(15) Accordingly, the input interface **204** may facilitate the receipt by the speech processing system **100** of the necessary data to initiate the real-time accent mimicking process described and illustrated herein. The input interface **204** may be the initial point of interaction between a user (e.g., a user computing device) or external systems and the speech processing system **100**. The input data provided through the input interface **204** may serve as the foundation for subsequent processing and analysis within the speech processing system **100**, as described and illustrated in detail below.

(16) The accent analysis module **200** is configured to analyze input speech from a first user (also referred to herein as first input speech) using machine learning model(s). In some examples, the accent analysis module **200** leverages pre-trained machine learning models to analyze captured first input speech and identify accent-specific features. The machine learning models in this example are trained on diverse speech datasets encompassing a wide range of accents and are adept at recognizing characteristics that distinguish one accent from another.

(17) The analysis by the accent analysis module **200** in some examples focuses on extracting key accent features including pitch contours or variation in pitch throughout the speech, intonation patterns including the rise and fall of pitch at the ends of phrases and sentences, and/or phoneme pronunciations or unique production of phonemes in different accents. These accent features extracted by accent analysis module **200** form a critical component for mimicking a first user's accent in a second user's speech (also referred to herein as second input speech), as explained in more detail below.

(18) The feature extraction module **212** is configured to extract linguistic features, prosodic features (e.g., pitch and timbre), and/or global speaker characteristics from the first input speech. The global speaker characteristics can include vocal timbre, speech rate, articulation style, pitch range, rhythm patterns, and/or accent-specific characteristics. The vocal timbre in some examples

is, the unique tonal quality of the speaker's voice, which can differentiate one speaker from another even when saying the same words. For example, a speaker with a warm, resonant timbre versus a speaker with a sharp, nasal timbre.

(19) The speech rate in some examples is the typical speed at which a speaker delivers speech. For instance, a speaker from a fast-paced linguistic environment may average **200** words per minute, while a speaker from a slower-paced environment may average **120** words per minute.

(20) The articulation style is the degree of clarity or slurring in a speaker's pronunciation. For example, some speakers enunciate every syllable clearly, while others may merge sounds, such as saying “gonna” instead of “going to.” The pitch range is the range of frequencies commonly used by a speaker. A speaker might naturally use a high-pitched voice with variations between 200-300 Hz, while another might operate in a low-pitched range, varying between 100-150 Hz.

(21) The rhythm patterns refer to the regularity and pattern of pauses, stress, and emphasis in a speaker's speech. For example, a speaker may consistently place emphasis on the first syllable of multisyllabic words or insert long pauses between sentences as part of their natural speaking style. The accent-specific characteristics in some examples include regional or cultural markers that define the speaker's accent. For instance, the tendency to roll the “r” sound in some accents or to flatten certain vowel sounds.

(22) The accent translation module **206** is configured to translate the linguistic features extracted by the feature extraction module **212** from a second accent (of the second input speech) to a first accent (of the first input speech). The first accent in some examples represents the first user's unique way of pronouncing words and structuring sentences.

(23) The synthesizer module **210** is configured to combine the extracted accent features, translated linguistic features, extracted prosodic features, and extracted global speaker characteristics to generate a modified version of the second input speech. The natural speech preservation module **202** is configured to understand the second user's natural speech characteristic(s) and substantially maintain the second user's natural voice during the modification of the second input speech.

(24) In some examples, the natural speech preservation module **202** employs techniques such as a mel frequency cepstral coefficient (MFCC) analysis, to extract a unique fingerprint of a second user's voice, and/or speaker identity encoding, to encode speaker-specific voice characteristics. These techniques are incorporated into the modified version of the second input speech generated using the synthesizer module **210** to allow the speech processing system **100** to maintain a natural sound throughout the process of modifying the second input speech. Thus, the natural speech preservation module **202** advantageously ensures the second user's speech substantially retains its natural quality while mimicking the accent of the first user represented within the first input speech.

(25) The output module **216** optionally facilitates adjustment of speech characteristics, such as speech rate, pitch, or gender, to further customize the representation of the modified version of the second input speech based on user preferences or application requirements, for example. The output module **216** optionally utilizes a vocoder to deliver a seamless and intelligible speech output that reflects the modified version of the second input speech with mimicked accent features. For example, by leveraging the advanced speech techniques described herein, the output module **216** may provide, in real-time or on-demand, a relatively accurate representation of second input speech from a second user in an accent that more closely corresponds to that of a first user.

(26) Referring now to FIG. **3**, a flow diagram of an exemplary method **300** for real-time accent mimicking is illustrated. In some examples, the method **300** may be implemented as a software application (e.g., software **116** executed by the central processing unit **102**) or a module within a larger system that includes the speech processing system **100**. The software application or module may receive input audio data, perform accent mimicking operations, and provide an output speech in real-time, as explained in detail below.

(27) Accordingly, in some examples, the steps **302-312** illustrated in FIG. **3** operate on the same

device (e.g., a first user computing device and, in other examples, a subset of the steps **302-312** (e.g., the accent translation of step **308** and/or synthesis of step **310**) may be executed on a second user computing device or a remote cloud server device, for example. Thus, in the former examples, no external processing is required. However, in the latter examples, the input speech can be transmitted from a first user computing device to a second user computing device, where the accent transformation occurs. The second user computing device in these examples then returns the transformed speech data to the first user computing device or directly outputs the speech to the target listener. Other permutations can also be used in other examples.

(28) Accordingly, in step **302** in some examples, the speech processing system **100** executing at a second user computing device, which may be remotely connected via communication networks to a first user computing device, receives second input audio data (e.g., via microphone and an audio interface) and extracts linguistic features (e.g., phonemes, syllables, word stress, speech rate, and/or pronunciation patterns) from second input speech represented by the second input audio data.

(29) The second input speech is associated with a second user of the second user computing device and a second accent of the second user. Optionally in parallel, the speech processing system **100** extracts prosodic features (e.g., pitch and timbre) from the second input speech in step **304** and global speech characteristics from the second input speech in step **306**.

(30) In step **308**, the speech processing system **100** translates the linguistic features extracted in step **302** from a second accent associated with the second user to a first accent associated with a first user of the first user computing device. The translation is facilitated by previously obtained accent-specific features. For example, the accent-specific features can be extracted by another speech processing system **100** executed at the first user computing device. The accent-specific features associated with the first accent are captured based on an analysis of first audio data representing first input speech by the first user. The analysis can leverage machine learning model(s) and the accent-specific features can include pitch contours, intonation patterns, and/or phoneme pronunciations, for example, although other accent-specific features can also be used in other examples.

(31) In step **310**, the speech processing system **100** combines the translated linguistic features generated in step **308**, the prosodic features extracted in step **304**, and the global speech characteristics extracted in step **306** to generate a modified version of the second input speech. In some examples, the synthesis in step **310** leverages a unique fingerprint of the second user's voice and/or encoded speaker-specific voice characteristics of the second input speech to thereby substantially maintain the second user's natural voice in the modified second input speech that mimics the first user's first accent.

(32) Accordingly, the synthesis in step **310** modifies the second input speech to mimic the first user's accent while preserving the natural voice characteristics of the second user. In some examples, this is achieved by leveraging a unique voice fingerprint of the second user and/or encoding speaker-specific characteristics, which may include pitch, timbre, and/or rhythm, for example, to ensure the second user's natural voice is substantially maintained. In this way, the modified second input speech retains the second user's identity (or natural voice characteristics), but with the accent of the first user's speech.

(33) In step **312**, the speech processing system uses a vocoder to turn the acoustic features of the modified version of the second input speech into output audio data and associated output speech. The output audio data and/or output speech can be sent from the second user computing device via one or more communication networks to the first user device for output via an output audio device (e.g., audio output device **130**) of the first user computing device.

(34) In other examples, the input audio data and/or input speech can be sent from the second user computing device via one or more communication networks to the first user computing device and the process illustrated in FIG. 3 can be performed by the speech processing system **100** executed at the first user computing device with the output speech output via an output audio device (e.g.,

audio output device **130**) of the first user computing device. Thus, any of the steps **302-312** can be executed on either of the first or second user computing device in some examples.

(35) Referring now to FIG. **4**, a flowchart of an exemplary method **400** for real-time accent mimicking is illustrated. In some examples, the method **400** may be implemented as a software application (e.g., software **116** executed by the central processing unit **102**) or a module within a larger system. The software application or module may receive input audio data, perform accent mimicking operations, and provide an output speech in real-time, as explained in detail below

(36) In step **402** in some examples, the speech processing system **100** receives first input speech associated with a first accent from a first user. The first input speech can be represented by first input audio data obtained via the audio interface **126** and a microphone **128**, for example, although the first input speech can also be obtained over one or more communication networks from another computing device in other examples.

(37) In step **404**, the speech processing system **100** analyzes and/or categorizes the first input speech using one or more machine learning models that are trained to recognize accent features that distinguish one accent from another accent. For example, the speech processing system **100** in step **404** may apply the machine learning models to distinguish features such as phonetic variations (e.g., how the speaker produces sounds that differ from another accent (e.g., vowel shifts, consonant articulation)), rhythm and stress patterns as different accents can involve varied speech rhythms and emphasis on certain syllables or words, and/or prosodic features (e.g., patterns in pitch, intonation, and/or cadence that help define accents).

(38) Based on the analysis in step **404**, the speech processing system **100** extracts the accent features from the first input speech in step **406**. In some examples, the extracted accent features include pitch contours or variation in pitch throughout the speech, intonation patterns including the rise and fall of pitch at the ends of phrases and sentences, and/or phoneme pronunciations or unique production of phonemes in different accents. These extracted accent features facilitate mimicking a first user's accent represented by the first input speech in a second user's speech.

(39) Additionally, in step **406** the speech processing system **100** can transform the identified features into a form that can be used for modifying the second input speech (e.g., to mimic the accent of the first input speech). This transformation can involve encoding the accent features in a way that preserves them while allowing for transformation in the next steps explained in detail below. Alternatively, or in combination, the transformation of step **406** can include normalization or standardization of the features to ensure that they are compatible with the processing pipeline of the speech processing system **100** as described and illustrated by way of the examples herein.

(40) In step **408**, the speech processing system **100** analyzes second input speech including a second accent from a second user using the same or one or more different machine learning model(s) as used in step **404** to generate characteristics specific to a natural voice of the second user. In some examples, steps **402-406** can occur at a first user computing device associated with the first user concurrently with steps **408-412** executed at a second user computing device. Thus, one or both of the first or second user computing devices can be separate instantiations of the speech processing system **100** in some examples, with the accent mimicking described and illustrated herein being performed in one or both directions between those user computing devices.

(41) The analysis in step **408** can include applying techniques such as MFCC or speaker identity encoding to the second input speech and/or extracting linguistic, prosodic, and/or global speaker characteristics from the second input speech. An MFCC analysis extracts a unique fingerprint of the second user's voice and a speaker identity encoding encodes speaker-specific voice characteristics of the second input speech. The fingerprint and/or encoding facilitate preservation of a natural sound of the second user's voice represented in the second input speech.

(42) The machine learning model(s) used by the speech processing system **100** in step **408** are designed to identify vocal traits that are distinct to the second user, including phonetic patterns (e.g., the specific sounds they produce), prosodic features (e.g., pitch, tempo, and/or stress

patterns), articulation styles, intonation patterns, and/or voice quality (e.g., including timbre and/or resonance). These features collectively contribute to what we recognize as the ‘natural voice’ of the second user.

(43) The purpose of the analysis in step **408** is to preserve natural voice identity and separate accent from identity. More specifically, the preservation of natural voice identity ensures that while the accent is being modified, the essential characteristics of the second user's natural voice remain intact, which is crucial in preventing the transformed speech from sounding artificial or mismatched with the second user's identity. The separation of accent from identity helps to distinguish between features that pertain to the second user's accent (i.e., the way speech sounds in terms of regional or social variations) and those that pertain to the speaker's inherent voice identity. This separate allows the speech processing system **100** to modify or adjust the accent without altering the second user's unique voice features.

(44) The machine learning model(s) used in step **408** can be trained on large datasets containing a variety of voices and accents, allowing them to recognize subtle differences in how speech is produced. These machine learning model(s) can be trained using supervised learning approaches, where a labeled dataset of various speakers' voice recordings is used to teach the speech processing system **100** how to distinguish between different accents and voice qualities. Other types of, and/or methods for training, the machine learning model(s) can also be used in other examples.

(45) Thus, the machine learning model(s) used in step **408** are configured to output extracted features that define the second user's voice, which can include pitch range and modulation (e.g., how the second user modulates their pitch during speech, which contributes to their unique voice signature), speech rate and rhythm (e.g., how fast or slow the second user speaks and the natural rhythm they follow, formant frequencies (e.g., the resonant frequencies of speech sounds, which help distinguish different accents), vocal timbre and resonance (e.g., the tonal quality of the voice that is unique to the speaker, which can be preserved during accent modification), and/or articulation patterns (e.g., the specific way the second user pronounces consonants and vowels, which is often influenced by their accent and voice mechanics). By analyzing the second input speech using the machine learning model(s), the speech processing system **100** ensures that the features of the second user's natural voice are accurately preserved while adjusting for the desired accent transformation. This process is fundamental to achieving a realistic and personalized speech output that mimics the first user's accent while maintaining the authenticity of the second user's voice.

(46) In step **410**, the speech processing system **100** modifies the second user's second input speech in real-time based on the accent features extracted in step **406** from the first input speech and the characteristics specific to the natural voice of the second user generated in step **408**. The synthesis in step **410** results in a modified version of the second input speech that preserves the natural quality of the second user's voice while mimicking the accent of the first user.

(47) In step **412**, the speech processing system delivers or outputs the modified version of the second input speech via output audio data, such as via the audio interface **126** and the audio output device **130**, for example. The methods and systems described and illustrated by way of the examples herein have many practical applications including in virtual conference platforms, language learning applications, accent reduction, immersive gaming and virtual reality experiences, and to provide accessibility for speech impairments.

(48) More specifically, the speech processing system **100** can be integrated into virtual conference platforms to enhance communication clarity by mimicking the accent of the primary speaker. In other examples, the speech processing system **100** can be used to provide real-time feedback to language learners by mimicking the accent of an instructor, helping learners improve their pronunciation and accent replication. The speech processing system **100** also can aid users with strong accents in adopting a listener's accent, enhancing communication clarity and reducing accent-related barriers.

(49) In yet other examples, the speech processing system **100** can be integrated into video games and virtual reality experiences to enhance immersion by adjusting a user's speech to match the accent(s) of in-game characters or environments. Additionally, the speech processing system **100** can be adapted to improve accessibility for users with speech impairments by enhancing clarity and reducing communication barriers. The advantages of this technology can be leveraged in many other use cases and types of deployments.

(50) Having thus described the basic concept of the invention, it will be rather apparent to those skilled in the art that the foregoing detailed disclosure is intended to be presented by way of example only and is not limiting. Various alterations, improvements, and modifications will occur and are intended for those skilled in the art, though not expressly stated herein. These alterations, improvements, and modifications are intended to be suggested hereby, and are within the spirit and scope of the invention. Additionally, the recited order of processing elements or sequences, or the use of numbers, letters, or other designations, therefore, is not intended to limit the claimed processes to any order except as may be specified in the claims. Accordingly, the invention is limited only by the following claims and equivalents thereto.

Claims

1. A speech processing system, comprising an audio interface coupled to a microphone and an audio output device, memory having instructions stored thereon, and one or more processors coupled to the memory and the audio interface and configured to execute the instructions to: apply one or more trained machine learning models to first input audio data obtained via the microphone and the audio interface to extract accent features of first input speech associated with a first accent of a first user; analyze obtained second input audio data associated with second input speech associated with a second accent of a second user to generate characteristics specific to a natural voice of the second user, wherein the generated characteristics correspond to vocal traits that are distinct to the second user and comprise one or more of a voice quality or one or more phonetic patterns, prosodic features, articulation styles, or intonation patterns; synthesize a modified version of the second input speech by modifying the obtained second input audio data based on the generated characteristics and the extracted accent features, wherein the modified version of the second input speech preserves aspects of the natural voice of the second user and mimics the first accent; and provide to the audio interface output audio data for output via the audio output device, wherein the output audio data is generated based on the modified version of the second input speech.

2. The speech processing system of claim 1, wherein the one or more processors are further configured to execute the instructions to extract from the first input audio data one or more prosodic features, linguistic features, or global speaker characteristics.

3. The speech processing system of claim 1, wherein the accent features comprise one or more pitch contours, other intonation patterns, or phoneme pronunciations and the pitch contours comprise variations in pitch throughout the first input speech, the other intonation patterns comprise the rise and fall of pitch at the ends of phrases or sentences, or the phoneme pronunciations comprise a unique production of phonemes in the first accent.

4. The speech processing system of claim 1, wherein the one or more processors are further configured to execute the instructions to apply a mel frequency cepstral coefficient (MFCC) analysis to extract a unique fingerprint of the voice of the second user, wherein the generated characteristics comprise the unique fingerprint.

5. The speech processing system of claim 1, wherein the one or more processors are further configured to execute the instructions to apply a speaker identity encoding technique to encode speaker-specific voice characteristics, wherein the generated characteristics comprise the speaker-specific voice characteristics.

6. The speech processing system of claim 1, wherein the one or more processors are further configured to execute the instructions to receive the second input audio data via one or more communication networks and from a user computing device that is remote from the speech processing system, wherein the second input audio data is captured at the user computing device.
7. A method for real-time accent mimicking, the method implemented by a speech processing system and comprising: applying one or more trained machine learning models to first input audio data to extract accent features of first input speech associated with a first accent of a first user; analyzing obtained second input audio data associated with second input speech associated with a second accent of a second user to generate characteristics specific to a natural voice of the second user, wherein the generated characteristics comprise a unique fingerprint of the voice of the second user; synthesizing a modified version of the second input speech by modifying the obtained second input audio data based on the generated characteristics and the extracted accent features; and providing output audio data generated based on the modified version of the second input speech.
8. The method of claim 7, wherein the modified version of the second input speech preserves aspects of the natural voice of the second user and mimics the first accent.
9. The method of claim 7, further comprising extracting from the first input audio data one or more prosodic features, linguistic features, or global speaker characteristics.
10. The method of claim 7, wherein the accent features comprise one or more pitch contours, intonation patterns, or phoneme pronunciations and the pitch contours comprise variations in pitch throughout the first input speech, the intonation patterns comprise the rise and fall of pitch at the ends of phrases or sentences, or the phoneme pronunciations comprise a unique production of phonemes in the first accent.
11. The method of claim 7, further comprising applying a mel frequency cepstral coefficient (MFCC) analysis to extract the unique fingerprint of the voice of the second user.
12. The method of claim 7, further comprising applying a speaker identity encoding technique to encode speaker-specific voice characteristics, wherein the generated characteristics comprise the speaker-specific voice characteristics.
13. The method of claim 7, further comprising receiving the second input audio data via one or more communication networks and from a user computing device that is remote from the speech processing system, wherein the second input audio data is captured at the user computing device.
14. A non-transitory computer-readable medium comprising instructions that, when executed by at least one processor, cause the at least one processor to: apply one or more trained machine learning models to first input audio data obtained via a microphone to extract accent features of first input speech associated with a first accent of a first user; analyze obtained second input audio data associated with second input speech associated with a second accent of a second user to generate characteristics specific to a natural voice of the second user, wherein the generated characteristics comprise speaker-specific voice characteristics; synthesize a modified version of the second input speech by modifying the obtained second input audio data based on the generated characteristics and the extracted accent features; and provide for output via an audio output device output audio data generated based on the modified version of the second input speech.
15. The non-transitory computer-readable medium of claim 14, wherein the modified version of the second input speech preserves aspects of the natural voice of the second user and mimics the first accent.
16. The non-transitory computer-readable medium of claim 14, wherein the instructions, when executed by the at least one processor further causes the at least one processor to extract from the first input audio data one or more prosodic features, linguistic features, or global speaker characteristics.
17. The non-transitory computer-readable medium of claim 14, wherein the accent features comprise one or more pitch contours, intonation patterns, or phoneme pronunciations and the pitch contours comprise variations in pitch throughout the first input speech, the intonation patterns

comprise the rise and fall of pitch at the ends of phrases or sentences, or the phoneme pronunciations comprise a unique production of phonemes in the first accent.

18. The non-transitory computer-readable medium of claim 14, wherein the instructions, when executed by the at least one processor further causes the at least one processor to apply a mel frequency cepstral coefficient (MFCC) analysis to extract a unique fingerprint of the voice of the second user, wherein the generated characteristics comprise the unique fingerprint.

19. The non-transitory computer-readable medium of claim 14, wherein the instructions, when executed by the at least one processor further causes the at least one processor to apply a speaker identity encoding technique to encode the speaker-specific voice characteristics.

20. The non-transitory computer-readable medium of claim 14, wherein the instructions, when executed by the at least one processor further causes the at least one processor to receive the second input audio data via one or more communication networks and from a user computing device that is remote from the speech processing system, wherein the second input audio data is captured at the user computing device.
